C

IB MATH: APPLICATIONS & INTERPRETATION • UNIT 8.01

PCA & Dimensionality Reduction

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

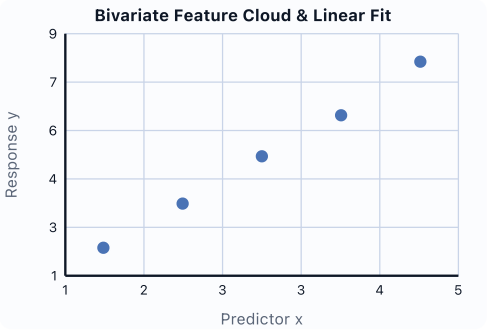
QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 D	Q2 A	Q3 A	Q4 A	Q5 A
Q6 A	Q7 A	Q8 A	Q9 D	Q10 B
Q11 A	Q12 C	Q13 D	Q14 A	Q15 C
Q16 B	Q17 D	Q18 C	Q19 A	Q20 D
Q21 A	Q22 B	Q23 B	Q24 C	Q25 D

PCA & Dimensionality Reduction Practice Exercises

1 MCQ

Principal Component Analysis (PCA) projects a dataset from R^{50} onto its top 6 principal components. What is the dimension of each transformed data point?



A 25

B 50

C 44

D 6 ✓

WORKED SOLUTION & VERIFICATION

PCA projection onto k eigenvectors transforms each sample vector from R^{D} to R^{k}. Here $k = 6$, so the reduced dimension is 6.

2 MCQ

A 3D dataset has covariance matrix eigenvalues $\lambda_1 = 62$, $\lambda_2 = 24$, and $\lambda_3 = 14$. What percentage of total variance is explained by the first principal component PC_1?

A 62.0% ✓

B 86.0%

C 24.0%

D 72.1%

WORKED SOLUTION & VERIFICATION

Total variance is $\sum \lambda_i = 62 + 24 + 14 = 100$. The proportion explained by PC_1 is $\frac{\lambda_1}{\sum \lambda_i} = \frac{62}{100} = 62.0\%$.

3 MCQ

A 3D dataset has covariance matrix eigenvalues $\lambda_1 = 64$, $\lambda_2 = 23$, and $\lambda_3 = 13$. What percentage of total variance is explained by the first principal component PC_1?

- **A** 64.0% ✓
- **B** 87.0%
- **C** 23.0%
- **D** 73.6%

WORKED SOLUTION & VERIFICATION

Total variance is $\sum \lambda_i = 64 + 23 + 13 = 100$. The proportion explained by PC_1 is $\frac{\lambda_1}{\sum \lambda_i} = \frac{64}{100} = 64.0\%$.

4 MCQ

A 3D dataset has covariance matrix eigenvalues $\lambda_1 = 66$, $\lambda_2 = 22$, and $\lambda_3 = 12$. What percentage of total variance is explained by the first principal component PC_1?

- **A** 66.0% ✓
- **B** 88.0%
- **C** 22.0%
- **D** 75.0%

WORKED SOLUTION & VERIFICATION

Total variance is $\sum \lambda_i = 66 + 22 + 12 = 100$. The proportion explained by PC_1 is $\frac{\lambda_1}{\sum \lambda_i} = \frac{66}{100} = 66.0\%$.

5 MCQ

A 3D dataset has covariance matrix eigenvalues $\lambda_1 = 68$, $\lambda_2 = 21$, and $\lambda_3 = 11$. What percentage of total variance is explained by the first principal component PC_1?

- **A** 68.0% ✓
- **B** 89.0%
- **C** 21.0%
- **D** 76.4%

WORKED SOLUTION & VERIFICATION

Total variance is $\sum \lambda_i = 68 + 21 + 11 = 100$. The proportion explained by PC_1 is $\frac{\lambda_1}{\sum \lambda_i} = \frac{68}{100} = 68.0\%$.

6 MCQ

A 3D dataset has covariance matrix eigenvalues $\lambda_1 = 70$, $\lambda_2 = 20$, and $\lambda_3 = 10$. What percentage of total variance is explained by the first principal component PC_1?

- **A** 70.0% ✓
- **B** 90.0%
- **C** 20.0%
- **D** 77.8%

WORKED SOLUTION & VERIFICATION

Total variance is $\sum \lambda_i = 70 + 20 + 10 = 100$. The proportion explained by PC_1 is $\frac{\lambda_1}{\sum \lambda_i} = \frac{70}{100} = 70.0\%$.

7 MCQ

A 3D dataset has covariance matrix eigenvalues $\lambda_1 = 72$, $\lambda_2 = 19$, and $\lambda_3 = 9$. What percentage of total variance is explained by the first principal component PC_1?

- **A** 72.0% ✓
- **B** 91.0%
- **C** 19.0%
- **D** 79.1%

WORKED SOLUTION & VERIFICATION

Total variance is $\sum \lambda_i = 72 + 19 + 9 = 100$. The proportion explained by PC_1 is $\frac{\lambda_1}{\sum \lambda_i} = \frac{72}{100} = 72.0\%$.

8 MCQ

A 3D dataset has covariance matrix eigenvalues $\lambda_1 = 74$, $\lambda_2 = 18$, and $\lambda_3 = 8$. What percentage of total variance is explained by the first principal component PC_1?

- **A** 74.0% ✓
- **B** 92.0%
- **C** 18.0%
- **D** 80.4%

WORKED SOLUTION & VERIFICATION

Total variance is $\sum \lambda_i = 74 + 18 + 8 = 100$. The proportion explained by PC_1 is $\frac{\lambda_1}{\sum \lambda_i} = \frac{74}{100} = 74.0\%$.

9 MCQ

Consider a 2D data point $\mathbf{v} = \begin{bmatrix} 3 \\ -4 \end{bmatrix}$. What is the squared Euclidean norm (or squared magnitude) of this vector?

- **A** 7
- **B** 9
- **C** 16
- **D** 25 ✓

WORKED SOLUTION & VERIFICATION

The squared Euclidean norm of a vector $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$ is given by $\|\mathbf{v}\|^2 = v_1^2 + v_2^2$. For the given vector $\mathbf{v} = \begin{bmatrix} 3 \\ -4 \end{bmatrix}$, the squared Euclidean norm is $(3)^2 + (-4)^2 = 9 + 16 = 25$.

10 MCQ

A binary classifier $p(t)$ produces the confusion matrix below with $\mathrm{TP} = 68$, $\mathrm{FP} = 10$, $\mathrm{FN} = 48$, and $\mathrm{TN} = 80$. Determine the total number of correct classifications $q = \mathrm{TP} + \mathrm{TN}$.

- **A** 143
- **B** 148 ✓
- **C** 165
- **D** 152

WORKED SOLUTION & VERIFICATION

Summing $\mathrm{TP} + \mathrm{TN} = 68 + 80 = 148$.

11 MCQ

Consider a dataset with two features, X_1 and X_2. The covariance matrix for this dataset is given by:

$$\Sigma = \begin{pmatrix} 10 & 5 \\ 5 & 20 \end{pmatrix}$$

What is the variance of the first feature, X_1?

- **A** 10 ✓
- **B** 5
- **C** 20
- **D** 25

WORKED SOLUTION & VERIFICATION

The covariance matrix Σ for a dataset with two features X_1 and X_2 is structured as follows:

$$\Sigma = \begin{pmatrix} Cov(X_1, X_1) & Cov(X_1, X_2) \\ Cov(X_2, X_1) & Cov(X_2, X_2) \end{pmatrix}$$

By definition, the variance of a random variable is the covariance of the variable with itself. Therefore, $Var(X_1) = Cov(X_1, X_1)$ and $Var(X_2) = Cov(X_2, X_2)$.
In the given covariance matrix:

$$\Sigma = \begin{pmatrix} 10 & 5 \\ 5 & 20 \end{pmatrix}$$

The element in the first row and first column represents $Cov(X_1, X_1)$, which is the variance of X_1.
Thus, the variance of X_1 is 10. The covariance between X_1 and X_2 is 5, and the variance of X_2 is 20.

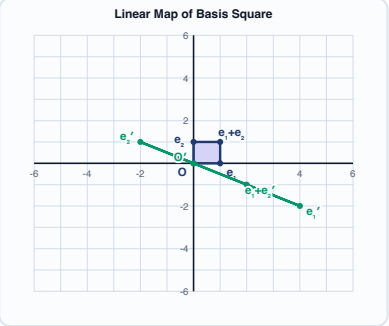
12 MCQ

Consider a dataset of observations $\{x_i\}_{i=1}^{n}$ where each x_i is a 2-dimensional vector. The sample covariance matrix for this dataset is given by:

$$S = \begin{pmatrix} 4 & -2 \\ -2 & 1 \end{pmatrix}$$

Principal Component Analysis (PCA) is applied to this dataset. The first principal component corresponds to the direction with the largest variance. Let the direction vector of the first principal component be $\mathbf{v}_1 = \begin{pmatrix} a \\ b \end{pmatrix}$.

The variance explained by this principal component is given by the corresponding eigenvalue λ_1.
If $\mathbf{v}_1$ is a unit vector, what is the value of $a^2 + b^2$?



A 2

B 4

C 1 ✓

D 5

WORKED SOLUTION & VERIFICATION

The first principal component direction vector $\mathbf{v}_1$ is an eigenvector of the covariance matrix S. The variance explained by this component is the corresponding eigenvalue λ_1. The problem states that $\mathbf{v}_1 = \begin{pmatrix} a \\ b \end{pmatrix}$ is a unit vector.

By definition, a unit vector has a magnitude of 1. The squared magnitude of a vector $\begin{pmatrix} a \\ b \end{pmatrix}$ is $a^2 + b^2$. Therefore, if $\mathbf{v}_1$ is a unit vector, its squared magnitude must be 1, which means $a^2 + b^2 = 1$. The specific values of a and b and the eigenvalues of S are not needed to answer this question, as it directly asks for the property of a unit vector.

13 MCQ

Consider a dataset of n data points, each residing in R^D. We are performing Principal Component Analysis (PCA) to reduce the dimensionality. The first three principal components capture 70% of the total variance, the first six capture 85%, and the first ten capture 92%. You are tasked with selecting a subset of principal components for a downstream machine learning model. If the model requires that at least 90% of the original variance be retained, and you want to use the minimum possible number of principal components, what is the dimensionality of the transformed data?

A D

B 3

C 6

D **10** ✓

WORKED SOLUTION & VERIFICATION

The problem requires us to find the minimum number of principal components such that the cumulative explained variance is at least 90%. The provided information states:

- The first 3 principal components capture 70% of the variance.
- The first 6 principal components capture 85% of the variance.
- The first 10 principal components capture 92% of the variance.

We need to find the smallest k such that the cumulative variance captured by the first k principal components is $\geq 90\%$.

Checking the given values:

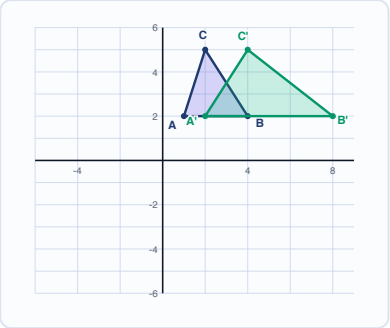
- For $k = 3$, variance captured = 70%, which is less than 90%.
- For $k = 6$, variance captured = 85%, which is less than 90%.
- For $k = 10$, variance captured = 92%, which is greater than or equal to 90%.

Therefore, the minimum number of principal components required to retain at least 90% of the variance is 10.

When PCA reduces the dimensionality of data from R^D to R^k using the top k principal components, the dimension of each transformed data point is k. In this case, $k = 10$.

14 MCQ

Consider a 2D dataset represented by the points $A = (1, 2)$, $B = (4, 2)$, and $C = (2, 5)$. If this dataset undergoes a linear transformation represented by the matrix $M = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$, what are the coordinates of the transformed points A', B', and C'?



- **A** $A' = (2, 2), B' = (8, 2), C' = (4, 5)$ ✓
- **B** $A' = (2, 4), B' = (8, 4), C' = (4, 10)$
- **C** $A' = (3, 2), B' = (6, 2), C' = (4, 7)$
- **D** $A' = (1, 4), B' = (4, 4), C' = (2, 10)$

WORKED SOLUTION & VERIFICATION

The transformation of a point $\mathbf{x}$ by a matrix M is given by $M\mathbf{x}$. We apply the matrix $M = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$ to each point:

For point $A = (1, 2)$, the transformed point A' is:
$\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \times 1 + 0 \times 2 \\ 0 \times 1 + 1 \times 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$ So, $A' = (2, 2)$.

For point $B = (4, 2)$, the transformed point B' is:
$\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 4 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$ So, $B' = (8, 2)$.

For point $C = (2, 5)$, the transformed point C' is:
$\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 5 \end{pmatrix} = \begin{pmatrix} 2 \times 2 + 0 \times 5 \\ 0 \times 2 + 1 \times 5 \end{pmatrix} = \begin{pmatrix} 4 \\ 5 \end{pmatrix}$ So, $C' = (4, 5)$.

Therefore, the transformed points are $A' = (2, 2)$, $B' = (8, 2)$, and $C' = (4, 5)$.

15 MCQ

In an applied problem, a 3D dataset has covariance matrix eigenvalues $\lambda_1 = 66$, $\lambda_2 = 22$, and $\lambda_3 = 12$. What percentage of total variance is explained by the first principal component PC_1?

- **A** 22.0%
- **B** 75.0%
- **C** 66.0% ✓
- **D** 88.0%

WORKED SOLUTION & VERIFICATION

Total variance is $\sum \lambda_i = 66 + 22 + 12 = 100$. The proportion explained by PC_1 is $\frac{\lambda_1}{\sum \lambda_i} = \frac{66}{100} = 66.0\%$.

16 MCQ

Suppose a 3D dataset has covariance matrix eigenvalues $\lambda_1 = 68$, $\lambda_2 = 21$, and $\lambda_3 = 11$. What percentage of total variance is explained by the first principal component PC_1?

- A 89.0%
- B 68.0% ✓
- C 21.0%
- D 76.4%

WORKED SOLUTION & VERIFICATION

Total variance is $\sum \lambda_i = 68 + 21 + 11 = 100$. The proportion explained by PC_1 is $\frac{\lambda_1}{\sum \lambda_i} = \frac{68}{100} = 68.0\%$.

17 MCQ

An observed 3D dataset has covariance matrix eigenvalues $\lambda_1 = 70$, $\lambda_2 = 20$, and $\lambda_3 = 10$. What percentage of total variance is explained by the first principal component PC_1?

- A 90.0%
- B 20.0%
- C 77.8%
- D 70.0% ✓

WORKED SOLUTION & VERIFICATION

Total variance is $\sum \lambda_i = 70 + 20 + 10 = 100$. The proportion explained by PC_1 is $\frac{\lambda_1}{\sum \lambda_i} = \frac{70}{100} = 70.0\%$.

18 MCQ

Consider a 3D dataset has covariance matrix eigenvalues $\lambda_1 = 72$, $\lambda_2 = 19$, and $\lambda_3 = 9$. What percentage of total variance is explained by the first principal component PC_1?

- A 91.0%
- B 19.0%
- C 72.0% ✓
- D 79.1%

WORKED SOLUTION & VERIFICATION

Total variance is $\sum \lambda_i = 72 + 19 + 9 = 100$. The proportion explained by PC_1 is $\frac{\lambda_1}{\sum \lambda_i} = \frac{72}{100} = 72.0\%$.

19 MCQ

In an applied problem, a 3D dataset has covariance matrix eigenvalues $\lambda_1 = 76$, $\lambda_2 = 17$, and $\lambda_3 = 7$. What percentage of total variance is explained by the first principal component PC_1?

- A 76.0% ✓
- B 81.7%
- C 93.0%
- D 17.0%

WORKED SOLUTION & VERIFICATION

Total variance is $\sum \lambda_i = 76 + 17 + 7 = 100$. The proportion explained by PC_1 is $\frac{\lambda_1}{\sum \lambda_i} = \frac{76}{100} = 76.0\%$.

20 MCQ

A centered feature matrix $\mathbf{X}$ has shape $N \times 14$. What is the size and symmetry property of the sample covariance matrix $\mathbf{\Sigma} = \frac{1}{N-1}\mathbf{X}^T\mathbf{X}$?

- A $N \times 14$ rectangular matrix
- B $N \times N$ symmetric matrix
- C 14×14 skew-symmetric matrix
- D 14×14 **symmetric matrix** ✓

WORKED SOLUTION & VERIFICATION

Since $\mathbf{X}$ has 14 columns, $\mathbf{X}^T$ is $14 \times N$ and $\mathbf{X}^T\mathbf{X}$ is 14×14. Furthermore, $(\mathbf{X}^T\mathbf{X})^T = \mathbf{X}^T(\mathbf{X}^T)^T = \mathbf{X}^T\mathbf{X}$, proving it is always symmetric.

21 MCQ

A centered feature matrix $\mathbf{X}$ has shape $N \times 18$. What is the size and symmetry property of the sample covariance matrix $\mathbf{\Sigma} = \frac{1}{N-1}\mathbf{X}^T\mathbf{X}$?

- A 18×18 **symmetric matrix** ✓
- B $N \times N$ symmetric matrix
- C 18×18 skew-symmetric matrix
- D $N \times 18$ rectangular matrix

WORKED SOLUTION & VERIFICATION

Since $\mathbf{X}$ has 18 columns, $\mathbf{X}^T$ is $18 \times N$ and $\mathbf{X}^T\mathbf{X}$ is 18×18. Furthermore, $(\mathbf{X}^T\mathbf{X})^T = \mathbf{X}^T(\mathbf{X}^T)^T = \mathbf{X}^T\mathbf{X}$, proving it is always symmetric.

22 MCQ

In an applied problem, a centered feature matrix $\mathbf{X}$ has shape $N \times 22$. What is the size and symmetry property of the sample covariance matrix $\mathbf{\Sigma} = \frac{1}{N-1}\mathbf{X}^T\mathbf{X}$?

- A $N \times N$ symmetric matrix
- B 22×22 **symmetric matrix** ✓
- C 22×22 skew-symmetric matrix
- D $N \times 22$ rectangular matrix

WORKED SOLUTION & VERIFICATION

Since $\mathbf{X}$ has 22 columns, $\mathbf{X}^T$ is $22 \times N$ and $\mathbf{X}^T\mathbf{X}$ is 22×22. Furthermore, $(\mathbf{X}^T\mathbf{X})^T = \mathbf{X}^T(\mathbf{X}^T)^T = \mathbf{X}^T\mathbf{X}$, proving it is always symmetric.

23 MCQ

Suppose a centered feature matrix $\mathbf{X}$ has shape $N \times 26$. What is the size and symmetry property of the sample covariance matrix $\mathbf{\Sigma} = \frac{1}{N-1}\mathbf{X}^T\mathbf{X}$?

- A $N \times N$ symmetric matrix
- B 26×26 **symmetric matrix** ✓
- C 26×26 skew-symmetric matrix
- D $N \times 26$ rectangular matrix

WORKED SOLUTION & VERIFICATION

Since $\mathbf{X}$ has 26 columns, $\mathbf{X}^T$ is $26 \times N$ and $\mathbf{X}^T\mathbf{X}$ is 26×26. Furthermore, $(\mathbf{X}^T\mathbf{X})^T = \mathbf{X}^T(\mathbf{X}^T)^T = \mathbf{X}^T\mathbf{X}$, proving it is always symmetric.

24 MCQ

An observed centered feature matrix $\mathbf{X}$ has shape $N \times 30$. What is the size and symmetry property of the sample covariance matrix $\mathbf{\Sigma} = \frac{1}{N-1}\mathbf{X}^T\mathbf{X}$?

A $N \times 30$ rectangular matrix

B $N \times N$ symmetric matrix

C 30×30 **symmetric matrix** ✓

D 30×30 skew-symmetric matrix

WORKED SOLUTION & VERIFICATION

Since $\mathbf{X}$ has 30 columns, $\mathbf{X}^T$ is $30 \times N$ and $\mathbf{X}^T\mathbf{X}$ is 30×30. Furthermore, $(\mathbf{X}^T\mathbf{X})^T = \mathbf{X}^T(\mathbf{X}^T)^T = \mathbf{X}^T\mathbf{X}$, proving it is always symmetric.

25 MCQ

Consider a centered feature matrix $\mathbf{X}$ has shape $N \times 34$. What is the size and symmetry property of the sample covariance matrix $\mathbf{\Sigma} = \frac{1}{N-1}\mathbf{X}^T\mathbf{X}$?

A $N \times N$ symmetric matrix

B 34×34 skew-symmetric matrix

C $N \times 34$ rectangular matrix

D 34×34 **symmetric matrix** ✓

WORKED SOLUTION & VERIFICATION

Since $\mathbf{X}$ has 34 columns, $\mathbf{X}^T$ is $34 \times N$ and $\mathbf{X}^T\mathbf{X}$ is 34×34. Furthermore, $(\mathbf{X}^T\mathbf{X})^T = \mathbf{X}^T(\mathbf{X}^T)^T = \mathbf{X}^T\mathbf{X}$, proving it is always symmetric.